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**Please circle your Class Section:**

C01 (Dr. Athar)

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**Name:** \_\_\_\_\_

**Student ID:** \_\_\_\_\_

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**ELEC ENG 3CL4 (Introduction to Control Systems)**  
**Final Examination**

Department of Electrical and Computer Engineering, McMaster University  
Instructors: Dr. Shahrukh Athar (C01) and Mr. Ayman Negm (C02), Winter 2023  
Monday, April 24, 2023, 12:30 PM to 3:00 PM (150 minutes)

**Examination Conditions** This test is conducted under standard examination conditions. In particular:

- You are not to communicate in any manner with any other student.
- You are not to look at the questions nor start writing until told to do so by an invigilator.
- When the invigilator announces that the test is over, you are to stop writing *immediately*.

The penalties for violating these or any other standard examination conditions will be severe, including a zero on the test and the commencement of formal disciplinary proceedings.

**Instructions**

- There are six (6) questions on this examination paper. Answer all six questions in the space provided in this exam paper. You may use the back of the pages if necessary.
- **When given permission to start writing, you must ensure that your copy of the exam paper is complete and bring any discrepancies to the attention of your exam invigilator.**
- Write your name and student number on the front page of this examination paper and on any additional booklet that you include with it. Encircle your section at the top of this page.
- Write your answers neatly and in a logical fashion. We may refuse to mark answers which are difficult to decipher.
- Direct answers without proper justifications will lead to reduced grade. However, justifications should be concise and to-the-point.
- You may write your test in pencil, but booklets written in pencil or with pages torn out will not be remarked.
- Closed-books, closed-notes.
- Aids permitted:

One double-sided (8.5 x 11) crib sheet with only handwritten notes (Not printed/photocopied).

McMaster Standard Calculator (Casio FX-991MS/MS+).

- Total Marks: 120 (mark allocations for questions/parts are shown within questions).

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| Q1 (20) | Q2 (20) | Q3 (20) | Q4 (20) | Q5 (20) | Q6 (20) | Total (120) |
|---------|---------|---------|---------|---------|---------|-------------|
|         |         |         |         |         |         |             |

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## Useful Information

- The solutions to the quadratic equation  $ax^2 + bx + c = 0$  are  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ .
- In the function  $f(t) = Ae^{-t/\tau}$  the term  $\tau$  is sometimes referred to as the time constant.
- The following Laplace transform pairs may be of use. Here  $u(t)$  is the unit step function

$$\begin{aligned}
 u(t) &\longleftrightarrow \frac{1}{s} \\
 e^{-at}u(t) &\longleftrightarrow \frac{1}{s+a} \\
 \sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{s^2 + \omega^2} \\
 \cos(\omega t)u(t) &\longleftrightarrow \frac{s}{s^2 + \omega^2} \\
 e^{-at} \cos(\omega t)u(t) &\longleftrightarrow \frac{s+a}{(s+a)^2 + \omega^2} \\
 e^{-at} \sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{(s+a)^2 + \omega^2} \\
 t^n u(t) &\longleftrightarrow \frac{n!}{s^{n+1}} \\
 \frac{df(t)}{dt} &\longleftrightarrow sF(s) - f(0^-) \\
 \frac{d^2f(t)}{dt^2} &\longleftrightarrow s^2F(s) - sf(0^-) - f'(0^-) \\
 \int_{0^-}^t f(\lambda) d\lambda &\longleftrightarrow \frac{F(s)}{s}
 \end{aligned}$$

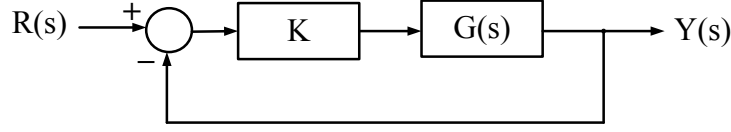
- A generic form for a rational Laplace Transform is  $G(s) = K_G \frac{\prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}$ , where the set of finite zeros is  $\{-z_i\}_{i=1}^M$  and the set of poles is  $\{-p_j\}_{j=1}^n$ .
- The final value theorem states that for a Laplace Transform  $Y(s)$  with all its poles in the left half plane, except for at most one pole at the origin,  $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$ .
- For a standard second-order system with a transfer function of the form  $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ , when  $0 < \zeta < 1$ , the poles are at positions  $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$ . The step response of such a system has the following properties:

- $y_{\text{step}}(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n\sqrt{1-\zeta^2} t + \theta)$ , where  $\theta = \cos^{-1}(\zeta)$ .
- 2% settling time,  $T_s \approx \frac{4}{\zeta\omega_n}$ .
- Percentage (maximum) overshoot,  $P.O. = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ . Hence,  $\zeta = \frac{-\ln(P.O./100)}{\sqrt{\pi^2 + \ln^2(P.O./100)}}$ .
- Time of peak,  $T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$ .
- For  $0.3 \leq \zeta \leq 0.8$ , the 10%-90% rise time is  $T_r \approx \frac{2.16\zeta + 0.6}{\omega_n}$ .

- Steady state error constants:

- Position error constant:  $K_{\text{posn}} = \lim_{s \rightarrow 0} G_c(s)G(s)$
- Velocity error constant:  $K_v = \lim_{s \rightarrow 0} sG_c(s)G(s)$
- Acceleration error constant:  $K_a = \lim_{s \rightarrow 0} s^2G_c(s)G(s)$

- Consider a feedback control system of the following form:

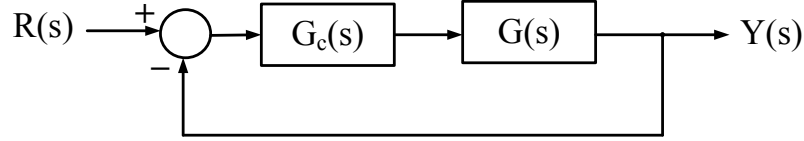


where  $K > 0$  and  $G(s)$  takes the form

$$G(s) = \frac{K_G \prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}, \text{ with } K_G > 0.$$

- The point  $s_0$  is on the root locus if  $\sum_{i=1}^M \angle(s_0 + z_i) - \sum_{j=1}^n \angle(s_0 + p_j) = 180^\circ + k360^\circ$  for some integer  $k$ .
- The value of the gain  $k$  such that  $s_0$  is a pole of the closed loop is  $K = \frac{\prod_{j=1}^n |s_0 + p_j|}{K_G \prod_{i=1}^M |s_0 + z_i|}$ .

- Consider the following closed-loop system (where the open-loop transfer function is  $L(s) = G_c(s)G(s)$ ):



- The gain margin is  $K_{gm} = \frac{1}{|L(j\omega_x)|} = \frac{1}{|G_c(j\omega_x)G(j\omega_x)|}$ , where  $\omega_x$  is the frequency at which  $\angle G_c(j\omega_x)G(j\omega_x) = -180^\circ$ .
- $K_{gm}$  (in dBs) =  $-20 \log_{10}(|L(j\omega_x)|)$
- The phase margin is  $\phi_{pm} = 180^\circ + \angle L(j\omega_c) = 180^\circ + \angle G_c(j\omega_c)G(j\omega_c)$ , where  $\omega_c$  (crossover frequency) is the frequency at which  $|G_c(j\omega_c)G(j\omega_c)| = 1$ .

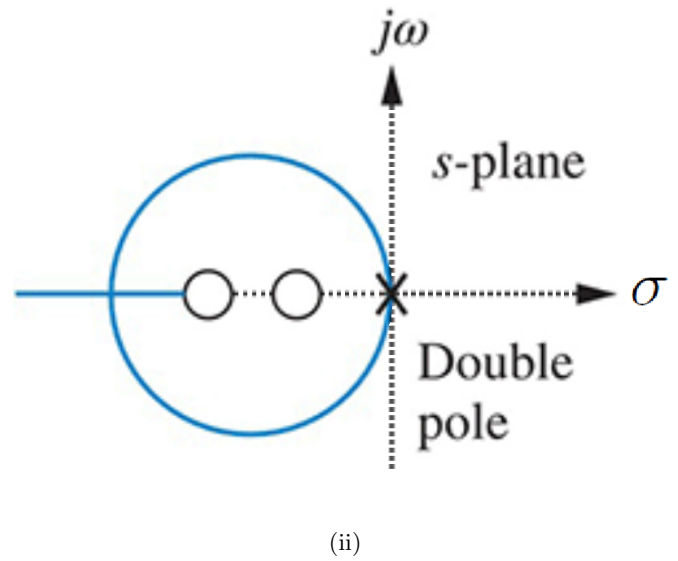
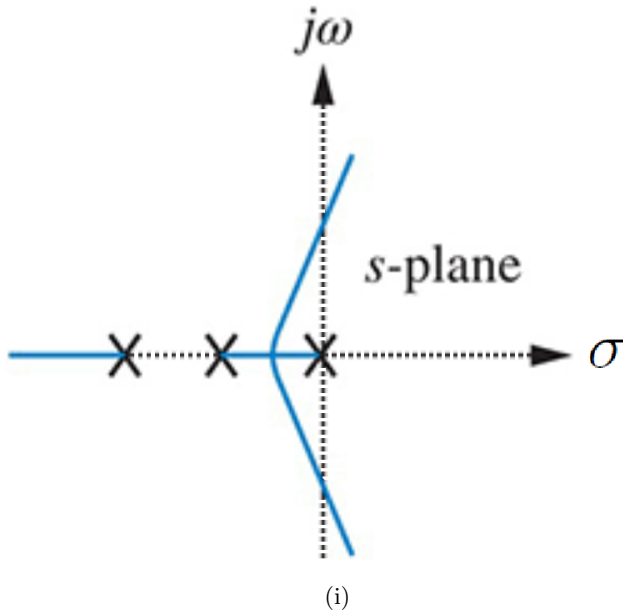
- In the frequency domain, for an open-loop system of the form  $L(s) = G_c(s)G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$ :

- The phase margin is:  $\phi_{pm} = \tan^{-1}\left(\frac{2}{\sqrt{(4+1/\zeta^4)^{1/2}-2}}\right)$ . If  $\phi_{pm}$  is measured in degrees, and if  $\zeta \leq \frac{1}{\sqrt{2}}$ , then  $\phi_{pm} \approx 100\zeta$ .
- The crossover frequency is:  $\omega_c = \omega_n(\sqrt{1+4\zeta^4} - 2\zeta^2)^{1/2}$ .
- The 2% settling time is:  $T_s = \frac{4f_c(\zeta)}{\omega_c}$  where  $f_c(\zeta) = (\sqrt{4+1/\zeta^4} - 2)^{1/2}$ .

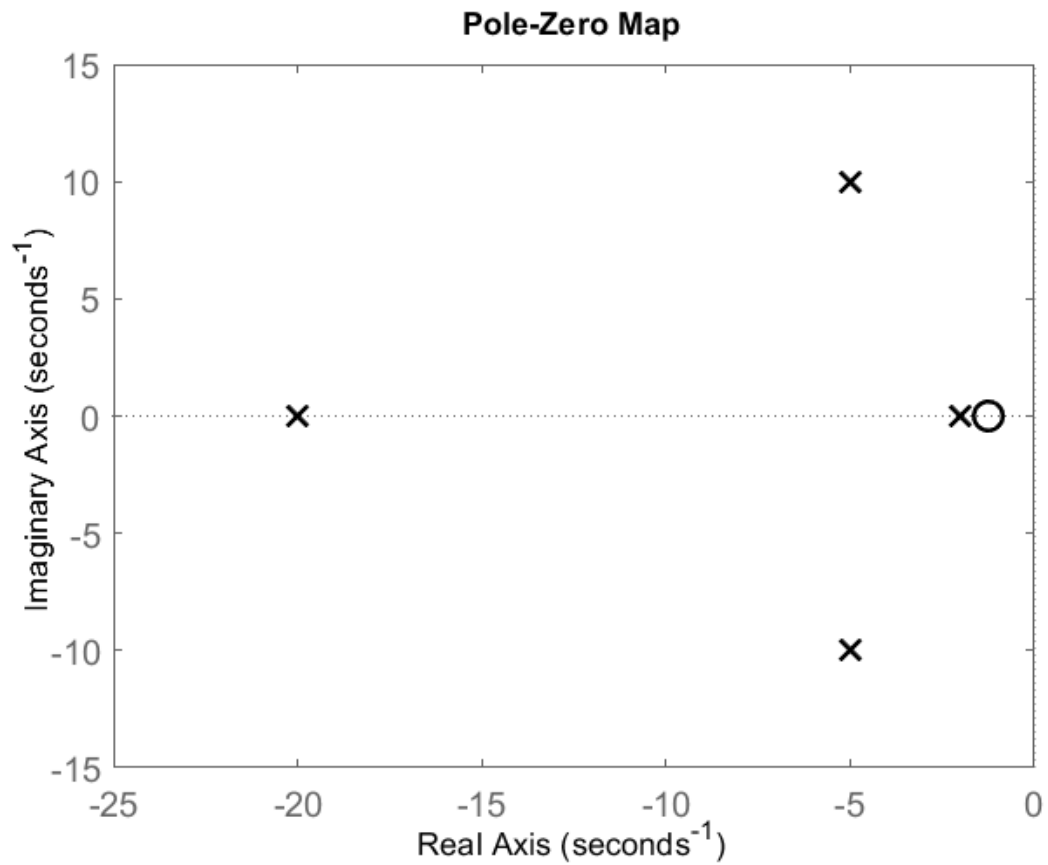
- A first order lead compensator has the form  $G_c(s) = \frac{K_c(s+z)}{s+p}$  with  $|z| < |p|$  and  $\alpha_{lead} = \frac{p}{z}$ . The peak of the phase response occurs at  $\omega_m = \sqrt{zp}$ .
- A first order lag compensator has the form  $G_c(s) = \frac{K_c(s+z)}{s+p}$  with  $|p| < |z|$  and  $\alpha_{lag} = \frac{z}{p}$ .

**Question 1 (20 marks).** This question is composed of five short parts (a-e). **Answers are expected to be short and concise.** We have provided more space than is necessary. Answer the following questions:

- a) (4 marks) You are given two sketches of root loci below (solid lines). Discuss whether each sketch can be a valid root locus or not.



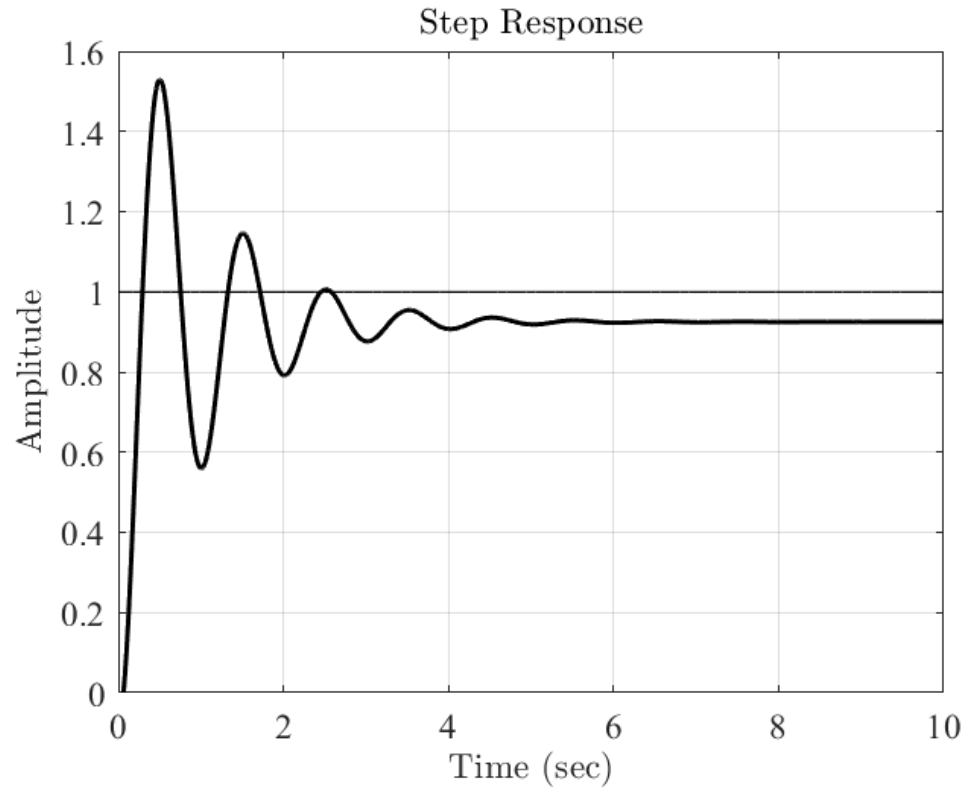
- b) (4 marks) The locations of the poles and zero of a fourth order closed-loop system are shown in the figure below. You are required to approximate the behavior of this system with either a first-order or a second-order system. Which one will you choose? Justify your answer.



c) (4 marks) Given a system with an open-loop transfer function of the form:

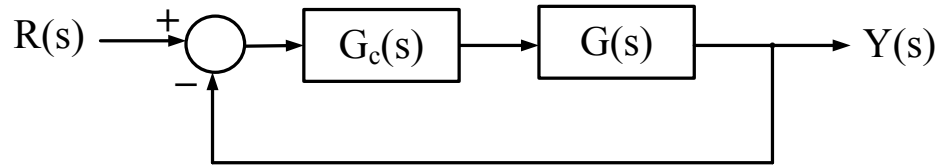
$$L(s) = \frac{K(s + a)}{s(s + b)(s + c)}$$

with all poles and zeros in the left half plane. Can such a system have a step response as depicted in the figure below? Assume that steady-state is achieved around  $t = 6$  sec. Justify your answer.



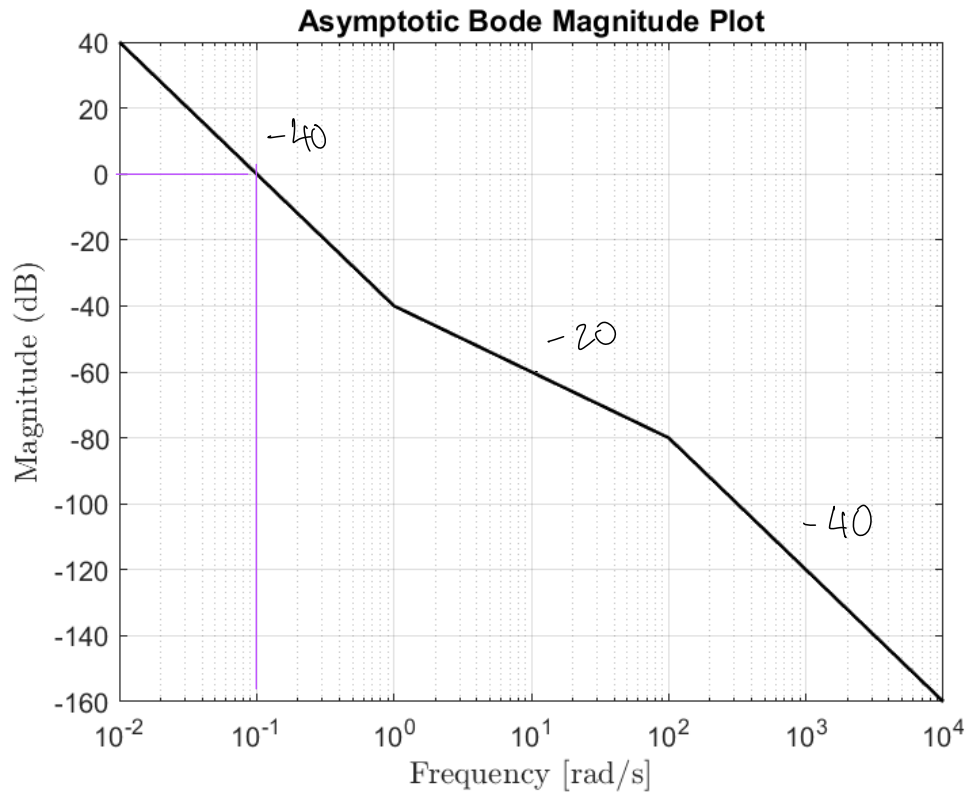
d) (4 marks) Consider the following feedback control system in which:

$$G(s) = \frac{4}{(s-2)(s+1)}, \quad G_c(s) = \frac{k(s-2)}{s+4}$$



Explain whether it is practical to use this controller to stabilize the system.

- e) (4 marks) Given below is the Bode magnitude diagram of the frequency response of an open-loop transfer function  $L(s)$ .

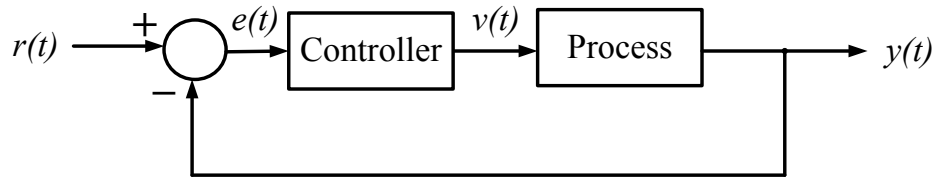


Just by observing the Bode diagram, answer the following:

- i What is the open-loop crossover frequency  $\omega_c$ ?
- ii When this open-loop transfer function is placed in a unity gain negative feedback loop, would the resulting system lead to zero, finite, or infinite steady state error to a ramp input? Justify your answer.

i)

**Question 2 (20 marks).**



Consider the closed-loop feedback control system above, in which the input-output relationship of the process is described by the differential equation:

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} = 2v(t)$$

For Parts (a-d) the input-output relationship of the controller is given by:

$$v(t) = K_{\text{amp}} e(t)$$

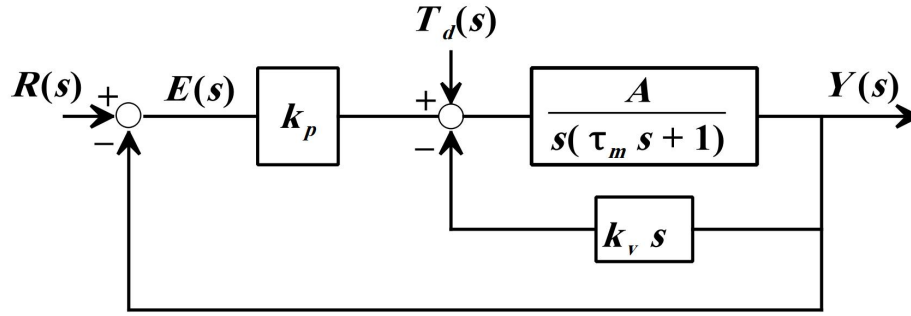
- a) **(5 marks)** Find the open-loop transfer function  $L(s)$ . What is the type number of the open-loop system?
- b) **(3 marks)** For what values of  $K_{\text{amp}}$  is the closed-loop stable?
- c) **(3 marks)** Find the set of values of  $K_{\text{amp}}$  for which the closed-loop system is underdamped.
- d) **(5 marks)** Sketch the root locus for this system and describe how your findings of Parts (b, c) are reflected in the root locus diagram.
- e) **(4 marks)** Now change the controller such that its input-output relationship is given by:

$$v(t) = e(t) + \int_0^t e(\tau) d\tau$$

Does this have an impact on the type number of the open-loop system? If so, how does this impact the steady-state error to a ramp input? Explain.

[Space for your answer to Question 2.]

**Question 3 (20 marks).**

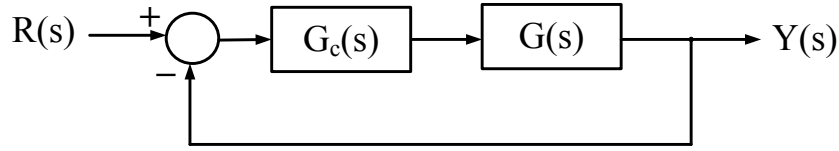


Consider the closed-loop feedback control system above, where  $R(s)$  is the reference input,  $T_d(s)$  is the disturbance,  $A = 100$ , and  $\tau_m = 0.125$  sec.

- (8 marks)** Derive the characteristic equation of the closed-loop system as a function of the system parameters  $k_p$  and  $k_v$ . (Show all steps)
- (6 marks)** Explain the effect of changing  $k_p$  and  $k_v$  on the overshoot and settling time of the unit-step response.
- (6 marks)** Calculate the steady-state output of the system when  $R(s) = 0$ ,  $T_d(s) = \frac{5}{s}$ ,  $k_p = 10$ , and  $k_v = 0.5$ .

[Space for your answer to Question 3.]

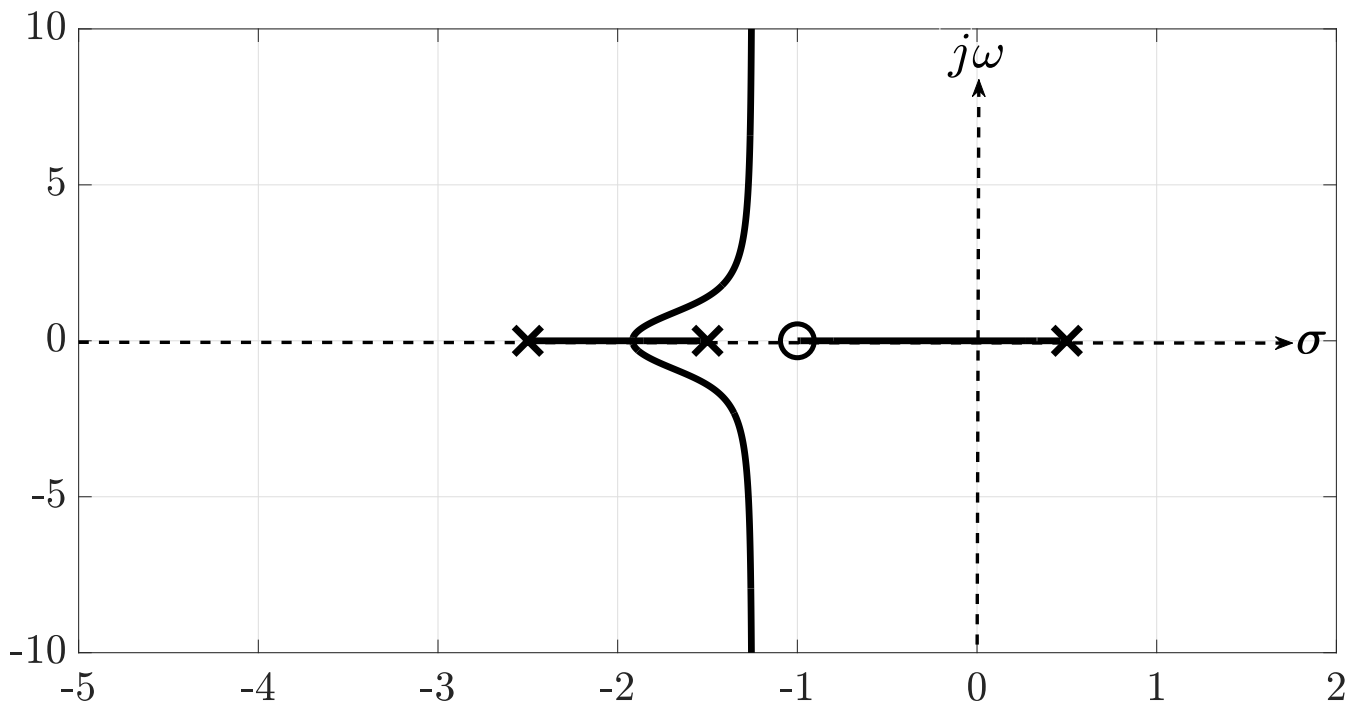
**Question 4 (20 marks).**



Consider the closed-loop feedback control system given in the figure above, in which

$$G(s) = \frac{\prod_{i=1}^M (s + z_i)}{\prod_{j=1}^n (s + p_j)}$$

where  $G(s)$  has no repeated poles or zeros. Given the following root locus plot of  $KG(s)$  as  $K$  goes from 0 to  $\infty$ ,



Design an appropriate compensator  $G_c(s)$  to achieve the following closed-loop step response characteristics:

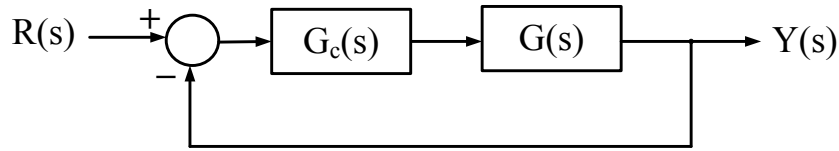
- A damping ratio of  $\zeta = 0.53$ .
- A 2% settling time of 1 second.

Clearly explain the principles that you use in each step of your design process.

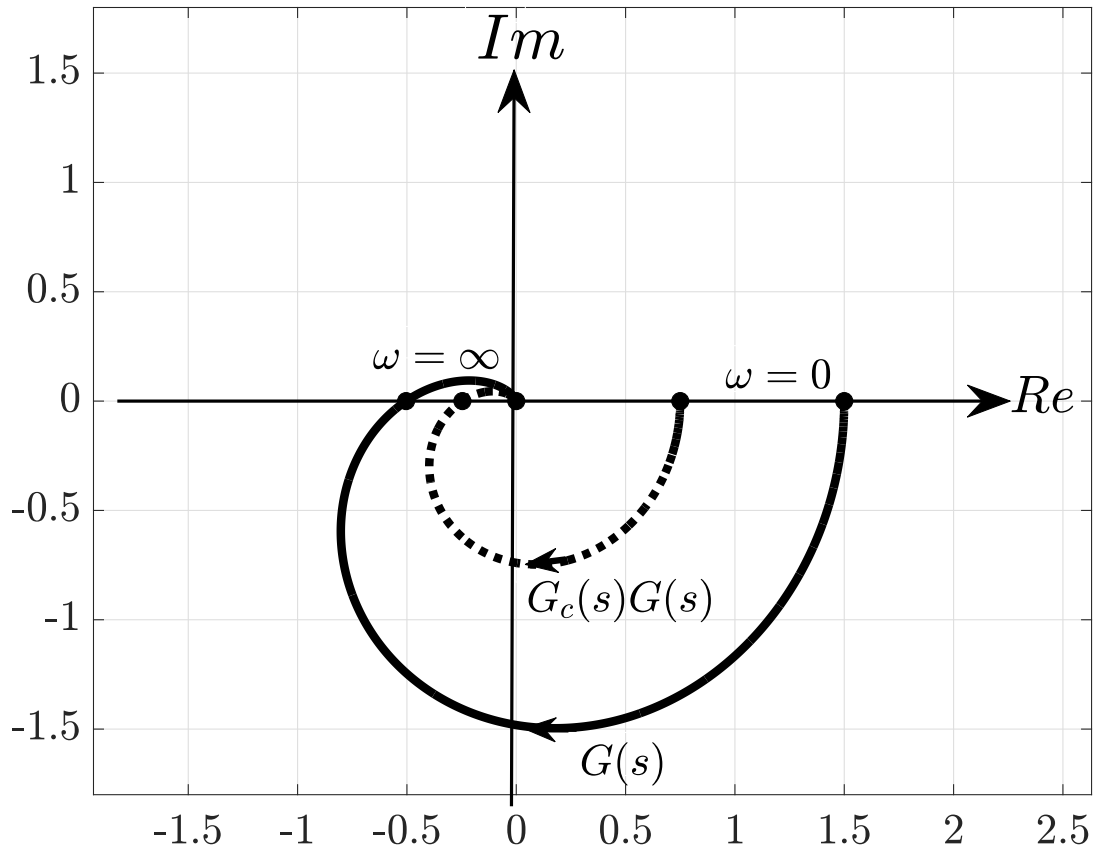
[Space for your answer to Question 4.]

[Space for your answer to Question 4.]

**Question 5 (20 marks).**

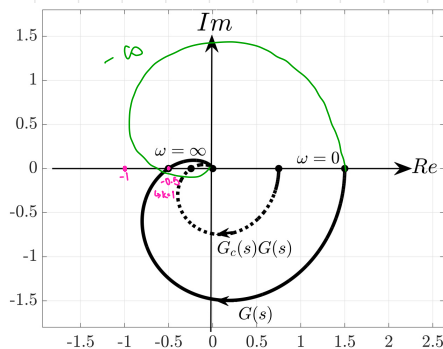


Consider the closed-loop feedback control system given in the figure above where  $G(s)$  has no poles in the right-half  $s$  - plane. The partially drawn Nyquist diagrams of  $G(s)$  (solid) and  $G_c(s)G(s)$  (dashed) are shown below, where the arrows indicate the direction of increasing frequency from 0 to  $\infty$ .



[Note that the  $G(s)$  curve applies to all five parts while the  $G_c(s)G(s)$  curve applies to Part (e) only.]

- (2 marks)** Find the magnitude and phase of  $G(s)$ , at  $\omega = 0$ .
- (2 marks)** Find the magnitude and phase of  $G(s)$  as  $\omega \rightarrow \infty$ .
- (2 marks)** Identify the type number of  $G(s)$ . Justify your answer.
- (8 marks)** Apply Nyquist Stability Criterion to find the maximum value of a proportional controller  $K$  such that the closed-loop is stable. (Show all steps)
- (6 marks)** Discuss the effect of  $G_c(s)$  on the system performance in terms of stability, phase-margin, and steady-state error to a step input.



- (2 marks) Find the magnitude and phase of  $G(s)$ , at  $\omega = 0$ .
- (2 marks) Find the magnitude and phase of  $G(s)$  as  $\omega \rightarrow \infty$ .
- (2 marks) Identify the type number of  $G(s)$ . Justify your answer.
- (8 marks) Apply Nyquist Stability Criterion to find the maximum value of a proportional controller  $K$  such that the closed-loop is stable. (Show all steps)
- (6 marks) Discuss the effect of  $G_c(s)$  on the system performance in terms of stability, phase-margin, and steady-state error to a step input.

a)  $\omega = 0$  :  $|G(s)| = 1.5$   $\angle G(s) = 0$   
 b)  $\omega \rightarrow \infty$  :  $|G(s)| = 0$   $\angle G(s) = 90^\circ \rightarrow \text{ZOOM in}$

c) type # @ low freq  $\rightarrow \omega = 0$

$$G(0) = \lim_{s \rightarrow 0} \frac{s+z}{s^n(s+p)} = \infty \rightarrow \text{for type } n, n \geq 1$$

$$G(0) = \lim_{s \rightarrow 0} \frac{s+z}{s+p} = \frac{z}{p} = \# \rightarrow G(0) = 1.5 \rightarrow \text{type } 0$$

type 0:  $\theta = 0^\circ$   
 \* type 1:  $\theta = 90^\circ$   
 type 2:  $\theta = 180^\circ$   
 type n:  $\theta = 90^\circ n$

d) other half just reflect across Re. axis  
 $\rightarrow -\infty \rightarrow \text{complex conj} \rightarrow \text{reflect across Re. axis}$

• nyquist theorem: # c.l. poles in RHP =  $P_L + N_L$

$P_L \rightarrow \text{open loop pole}$

$N_L \rightarrow \# \text{ of encirclements around pt @ } -1$

$$-0.5K > -1$$

$$0.5K < 1$$

$$K < \frac{1}{0.5}$$

$$K < 2$$

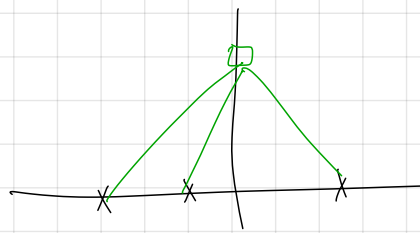
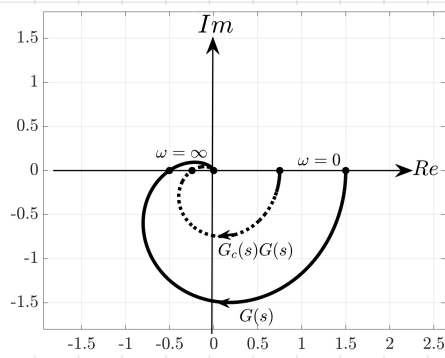
e) • stability :  $L(s)$  makes more stable  $\rightarrow$  it further away from -1 pt than  $G(s)$

•  $\phi_{PM}$  : for no phase  $L(s)$  of will it make it reach -1

$\phi_{PM} = \infty \rightarrow \text{better off}$

•  $e_{ss}$  :  $e_{ss} = \frac{A}{1+K_{pos}} \rightarrow K_{pos} \text{ smaller w/ } L(s)$   
 $e_{ss} \text{ gets bigger w/ } L(s)$

$\therefore \text{worse w/ } L(s)$



$$G(s) = \frac{1}{(s+a)(s+b)(s+c)}$$

$$\angle G(j\omega) = \angle \frac{1}{(j\omega + a)(j\omega + b)(j\omega + c)}$$

$$= 0 - \angle(j\omega + a) - \angle(j\omega + b) - \angle(j\omega + c)$$

$$= 0 - \angle(j\omega + a) - \angle(j\omega + b) - \angle(j\omega + c)$$

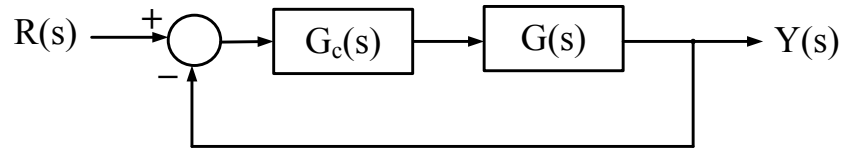
$$= -\theta_1 - \theta_2 - \theta_3$$

$$270^\circ =$$

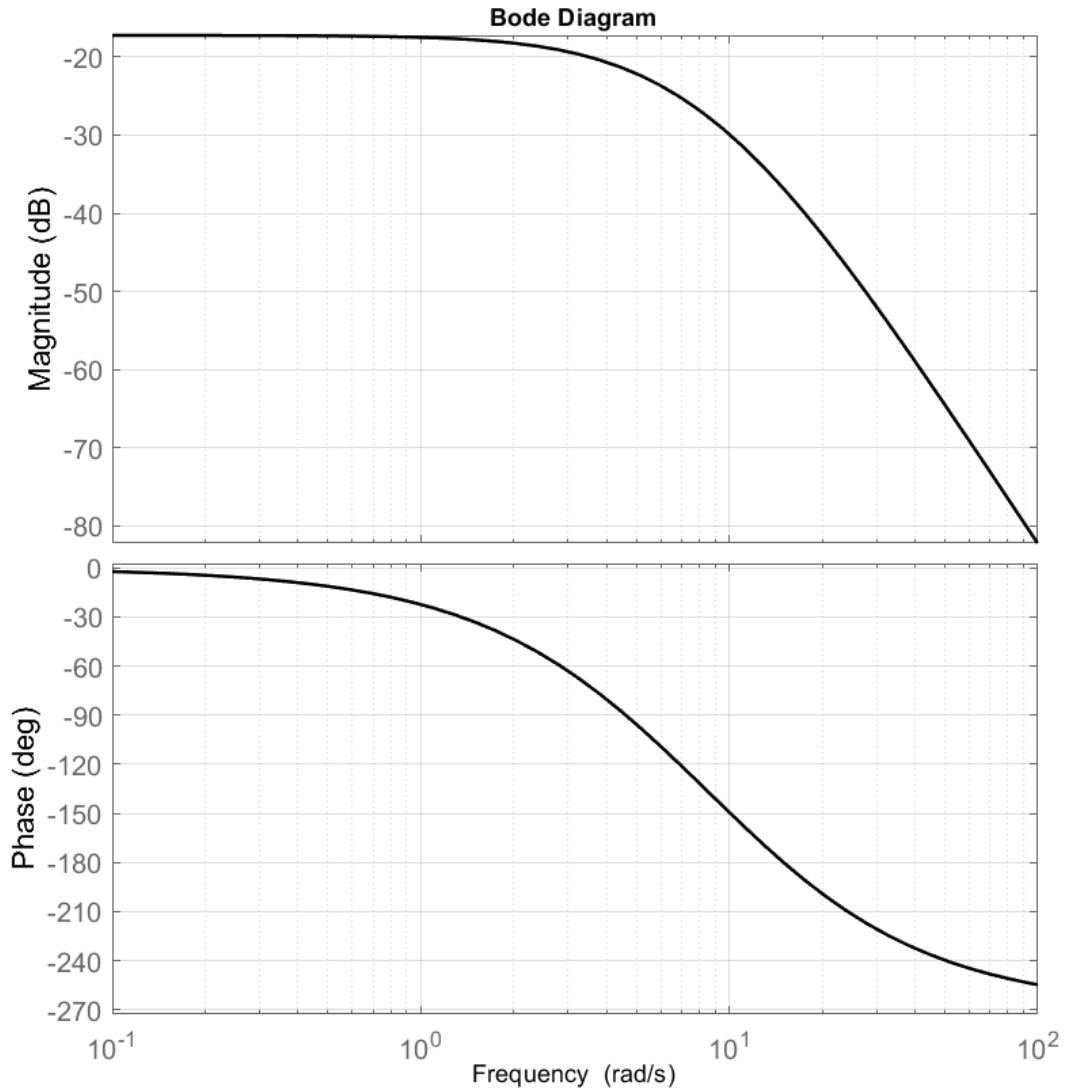
↳ each pole contributes  $90^\circ$

[Space for your answer to Question 5.]

**Question 6 (20 marks).** Consider the following closed-loop feedback control system:



The frequency response of the stable plant  $G(s)$  is given in the following diagram:

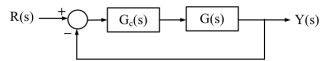


Design the controller  $G_c(s)$  so that all of the following conditions are met:

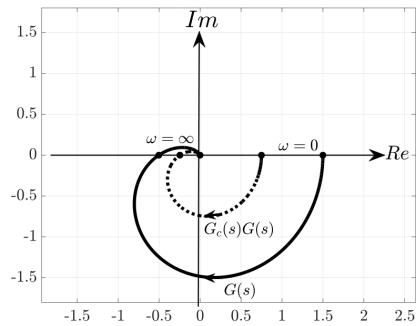
- The open-loop crossover frequency is  $\omega_c = 10$  rad/sec.
- The dominant closed-loop poles have a damping ratio of  $\zeta = 0.6971$ .
- The steady-state error to a step input is less than 2% of the size of the step.

Clearly explain the principles that you use in each step of your design process.

Question 5 (20 marks).



Consider the closed-loop feedback control system given in the figure above where  $G(s)$  has no poles in the right-half  $s - plane$ . The partially drawn Nyquist diagrams of  $G(s)$  (solid) and  $G_c(s)G(s)$  (dashed) are shown below, where the arrows indicate the direction of increasing frequency from 0 to  $\infty$ .



[Note that the  $G(s)$  curve applies to all five parts while the  $G_c(s)G(s)$  curve applies to Part (e) only.]

- a) **(2 marks)** Find the magnitude and phase of  $G(s)$ , at  $\omega = 0$ .
- b) **(2 marks)** Find the magnitude and phase of  $G(s)$  as  $\omega \rightarrow \infty$ .
- c) **(2 marks)** Identify the type number of  $G(s)$ . Justify your answer.
- d) **(8 marks)** Apply Nyquist Stability Criterion to find the maximum value of a proportional controller  $K$  such that the closed-loop is stable. (Show all steps)
- e) **(6 marks)** Discuss the effect of  $G_c(s)$  on the system performance in terms of stability, phase-margin, and steady-state error to a step input.

[Space for your answer to Question 6.]

[Space for your answer to Question 6.]